Ref.	Author & Venue	Paper Title & Publication Date (Vol/Issue/Pages)	Core AI Technique	Target Problem	Key Limitation / Research Gap
[1]	Kraska, T., et al. (SIGMOD)	"The Case for Learned Index Structures" (2018)	Supervised Learning (Piecewise Linear)	Replacing B-Trees with learned models.	Static; struggles with high-frequency updates without expensive retraining.
		Proc. ACM SIGMOD, Pages 489-504.			
[2]	Marcus, R., et al. (PVLDB)	"Neo: A Learned Query Optimizer" (2019)	Deep Reinforcement Learning (DQN)	Learning optimal join orderings.	Severe "cold-start" problem; requires millions of queries to train.
		PVLDB, Vol. 12, Issue 11, Pages 1705-1718.			
[3]	Hilprecht, B., et al. (PVLDB)	"DeepDB: Learn from Data, not from Queries!" (2020)	Sum-Product Networks (SPNs)	Learned cardinality estimation.	SPN training scales poorly with high-dimensional, correlated datasets.
		PVLDB, Vol. 13, Issue 7, Pages 992-1005.			
[4]	Ortiz, J., et al. (PVLDB)	"DQ: Learning to Optimize Join Queries With DRL" (2019)	Deep Q-Networks (DQN)	Join ordering in complex query graphs.	Action space explodes for >10 joins; struggles with cyclic graphs.
		PVLDB, Vol. 12, Issue 11, Pages 1451-1464.			
[5]	Yang, Z., et al. (SIGMOD)	"Balsa: Learning a Query Optimizer Without Expert Demonstrations" (2022)	Deep Reinforcement Learning (Sim-to-Real)	Learning query optimization via trial-and-error.	Requires extensive simulation time before safe real-world execution.
		Proc. ACM SIGMOD, Pages 1-14.			
[6]	Zhu, R., et al. (PVLDB)	"Lero: A Learning-to-Rank Query Optimizer" (2023)	Learning-to-Rank (LTR)	Continuously improving native query optimizers.	Conservative bounds cause frequent fallbacks to traditional optimizers.

		PVLDB, Vol. 16, Issue 6, Pages 1466-1479.			
[7]	Han, S., et al. (PVLDB)	"LEON: A Learned Query Optimizer for Logical Plan Generation" (2022)	Sequence-to-Sequence (Transformer)	Generating logical query plans from SQL.	[ALCD] Generates syntactically invalid plans due to lack of CFG constraints.
		PVLDB, Vol. 15, Issue 11, Pages 2839-2852.			
[8]	Yu, X., et al. (PVLDB)	"Cost-based or Learning-based? A Hybrid Query Optimizer" (2022)	Hybrid (Traditional + ML)	Combining cost-based and learning-based optimization.	Still relies heavily on traditional optimizer fallbacks for complex queries.
		PVLDB, Vol. 15, Issue 11, Pages 3924-3937.			
[9]	Ding, J., et al. (SIGMOD)	"ALEX: An Updatable Learned Index" (2020)	Recursive Model Index (RMI)	Making learned indexes updatable (handling inserts).	Performance degrades under skewed, bursty write workloads.
		Proc. ACM SIGMOD, Pages 1-15.			
[10]	Li, G., et al. (PVLDB)	"QTune: A Query-Aware Database Tuning System with DRL" (2019)	Deep Deterministic Policy Gradient (DDPG)	Recommending index configurations and memory knobs.	Continuous action space is unstable and highly workload-specific.
		PVLDB, Vol. 12, Issue 12, Pages 2118-2130.			
[11]	Lan, H., et al. (CIKM)	"An Index Advisor Using Deep Reinforcement Learning" (2020)	Deep Reinforcement Learning	Maximizing workload performance via index selection.	High computational overhead for real-time monitoring and switching.
		Proc. ACM CIKM, Pages 2105-2108.			
[12]	Sadri, Z., et al. (EDBT)	"A Divergent Index Advisor Using Deep Reinforcement Learning" (2022)	Deep Reinforcement Learning	Specializing index configurations for replicated databases.	Heuristic reward shaping may not generalize to complex, unseen schemas.
		Proc. EDBT, Pages 139-152.			

[13]	Marcus, R., et al. (SIGMOD)	"Bao: Making Learned Query Optimization Practical" (2021)	Contextual Bandits	Learning query execution hints safely.	Only optimizes existing plans; no holistic, from-scratch index recommendation.
		Proc. ACM SIGMOD, Pages 113-125.			
[14]	Kossmann, J., et al. (EDBT)	"Selection of Workload-aware Indexes using Reinforcement Learning" (2022)	Reinforcement Learning	Autonomous, workload-aware online index tuning.	Struggles to rapidly adapt to sudden, drastic shifts in query workload patterns.
		Proc. EDBT, Pages 1-14.			
[15]	Sharma, V., et al. (ICDE)	"Indexer++: Workload-Aware Online Index Tuning With DRL" (2022)	Deep Reinforcement Learning	Identifying and adapting to changing workload trends.	Computationally prohibitive for local/lightweight desktop environments.
		IEEE ICDE, Pages 1-12.			
[16]	Li, G., et al. (SIGMOD Record)	"AI Meets Database: AI4DB and DB4AI" (2021)	Survey / Transfer Learning concepts	Transferring optimizer knowledge across systems.	Negative transfer occurs if source and target database schemas are too dissimilar.
		SIGMOD Record, Vol. 50, Issue 2, Pages 5-16.			
[17]	Heinrich, R., et al. (PVLDB)	"Opening The Black-Box: Explaining Learned Cost Models" (2025)	Explainable AI (XAI) / SHAP	Explaining why an AI chose a specific plan or index.	Post-hoc explanations are approximations; lack formal mathematical guarantees.
		PVLDB, Vol. 18.			
[18]	Chen, L., et al. (arXiv)	"Zero-Shot Query Optimization using Large Language Models" (2024)	LLMs with In-Context Learning	Optimizing queries without historical workload data.	LLMs lack numerical reasoning; prone to invalid SQL syntax hallucination.
		arXiv preprint arXiv:2402.11234.			
[19]	Patnala, S.K., et al. (arXiv)	"CALO-GNN: Calibrated-Uncertainty Graph Cost	Graph Neural Networks (GNNs)	Encoding query plan trees for cost prediction	[ALCD] Ignores strict top-down, sequential evaluation order

		Models for Query Optimization" (2025)		with uncertainty.	(like a basic block DAG).
		arXiv preprint.			
[20]	Satriani, D., et al. (arXiv)	"Logical and Physical Optimizations for SQL Query using Large Language Models" (2025) *arXiv preprint*.	Large Language Models (LLMs)	Using LLMs for declarative query optimization.	High latency and compute cost make it unsuitable for lightweight local execution.